Azure Virtual Desktop — Deployment, Golden Image & FSLogix

Implementation Solution Guide

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Executive Summary — Azure Virtual Desktop: Deployment, Golden Image & FSLogix

Azure Solutions

Deploying Azure Virtual Desktop (AVD) by hand produces snowflake session hosts, roaming-profile pain, and a runaway bill. Every host is patched slightly differently, profiles do not follow users cleanly, and idle VMs run all night. This package builds AVD the engineered way: a **versioned golden image** produced by a repeatable build pipeline, session hosts deployed from that image by **infrastructure-as-code (Bicep)**, **FSLogix profile containers** on a correctly sized Azure Files or Azure NetApp Files share, **role-based access control** on the AVD objects, a **drain-mode patch workflow** for updates, and **autoscale scaling plans** that shut idle hosts down.

This is a complete, **deploy-it-today** package for the *build and operate* side of AVD, written for an administrator with no prior image-pipeline or IaC experience.

Scope boundary

This solution covers the **deployment, imaging, host-pool build, FSLogix storage, and scaling** side of AVD. It does **not** re-cover end-user client access and connectivity from Windows and macOS — that is the companion **Azure Virtual Desktop Remote Access (Windows & macOS Clients)** solution. Where the two meet (host pools, workspaces, FSLogix), this guide cross-references rather than duplicates.

What you get

- A plain-language **Solution Overview** of golden images, Azure Compute Gallery versioning, image-build pipelines, IaC session-host deployment, FSLogix storage sizing, RBAC, drain/patch, and scaling plans.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite specific to the build — a build/source VM, Azure Compute Gallery, FSLogix storage account, IaC tooling, and Azure roles.
- An **Implementation Guide** of precise, ordered steps: build and capture the golden image, version it in a gallery, deploy pilot session hosts from Bicep, wire up FSLogix, assign RBAC, apply a scaling plan, then expand.
- Five **automation artifacts** — read-only discovery, validation, and reporting scripts (PowerShell/Az), a **Bicep** session-host + scaling-plan template, and a gated drain/patch maintenance script.
- **Testing** and **Troubleshooting** guides for a confident rollout.

What this delivers

A repeatable, cost-controlled AVD estate: session hosts built from one **versioned image** in an Azure Compute Gallery, deployed by **Bicep** (not hand-built), profiles roamed by **FSLogix** on a right-sized share, updates rolled out with **drain mode**, and idle capacity removed by an **autoscale scaling plan** — proven on a pilot host pool before you scale.

The deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

Consistent golden images, roaming profiles, IaC-deployed hosts, and automatic cost control — an AVD estate a single administrator can build, version, and operate, proven on a pilot before production.

Effort

Roughly two to four days for a first pilot host pool (image build and capture is the longest single step), then hours per subsequent wave because the image, the Bicep template, and the scaling plan are reused.

1. Solution Overview — Azure Virtual Desktop: Deployment, Golden Image & FSLogix

What this solution is

This is the **build-and-operate** side of Azure Virtual Desktop (AVD). Where the companion *AVD Remote Access* solution covers how end users connect from Windows and macOS clients, this solution covers how you **produce a consistent golden image**, **deploy session hosts from it with code**, **store user profiles at scale with FSLogix**, **control who can administer the estate (RBAC)**, and **keep cost down with autoscale** — so the estate is repeatable instead of hand-built.

If you have not yet created any AVD host pool at all, deploy the pilot host pool described here; the *AVD Remote Access* solution then covers publishing it to end users and validating the Windows/macOS client experience. The two are designed to be used together without overlap.

How the technology works

1. A **golden image** is a single, fully configured Windows master (Windows 11 Enterprise multi-session, plus your agents, runtime, and line-of-business apps) that every session host is built from. Building once and cloning many is what makes hosts identical instead of snowflakes.

2. An **Azure Compute Gallery** (formerly Shared Image Gallery) stores that image as an **image definition** with numbered, immutable **image versions** (for example `1.0.0`, `1.1.0`). Because each version is immutable, you always know exactly what a host was built from, and you can replicate a version to multiple regions and roll back to a prior version.

3. An **image-build pipeline** turns "build a new image" into a repeatable job. It starts from a source (the Azure Marketplace base image or your previous gallery version), applies your customizations (Windows Update, app installs, FSLogix agent, optimizations), then captures the result as a **new image version**. This can be a documented manual capture, or automated with **Azure VM Image Builder** (Azure Image Builder / VM image templates) so the same definition rebuilds monthly.

4. **Session hosts** are the real Windows VMs that carry user sessions. Deploying them from **infrastructure-as-code (Bicep)** — referencing a specific gallery image version — means a host pool of ten or a hundred hosts is one reviewable, re-runnable template, not a hundred manual VM creations.

5. **FSLogix profile containers** store each user's Windows profile as a VHDX on a file share (**Azure Files** or **Azure NetApp Files**). On a pooled multi-session host pool, a user can land on any host and their profile, settings, and Outlook/Teams cache follow them. Correct **sizing, redundancy, and permissions** on that share is the single biggest driver of logon performance.

6. **Role-based access control (RBAC)** on the AVD objects (using the built-in *Desktop Virtualization* roles) governs who can operate the host pools, workspaces, and application groups without handing out broad subscription rights.

7. A **scaling plan** attached to the host pool ramps session hosts up before business hours and drains/deallocates them when idle, so you pay for compute only when people are working.

8. A **drain-and-patch workflow** takes hosts out of rotation gracefully (drain mode), updates them (or replaces them with a new image version), and returns them to service — so updates never disconnect an active user mid-session.

The standard deployment model

> ***Assess*** *your subscription, licensing, regions, and where profiles and images will live →* ***Configure*** *the Compute Gallery, build and version the golden image, and prepare FSLogix storage →* ***Pilot*** *a small host pool deployed from the image by Bicep, with FSLogix and a scaling plan →* ***Validate*** *image currency, host health, profile roaming, RBAC, and scaling against explicit pass/fail gates →* ***Expand*** *to production waves, reusing the same image, template, and scaling plan. The repeatable model:* ***Assess*** *→* ***Configure*** *→* ***Pilot*** *→* ***Validate*** *→* ***Expand.***

What you can control

• **Image contents and cadence** — what is baked into the golden image, and how often a new version is built.

• **Versioning and rollback** — immutable gallery versions and multi-region replication; roll a host pool back to a prior version if a new one regresses.

• **Deployment method** — session hosts deployed reproducibly from a Bicep template pinned to a specific image version.

• **Profile storage** — FSLogix on Azure Files vs Azure NetApp Files, share sizing, redundancy, and identity (Kerberos) model.

• **Administration** — least-privilege RBAC on the AVD objects using the built-in Desktop Virtualization roles.

• **Cost** — autoscale scaling plans, ramp schedules, and capacity thresholds.

• **Updates** — drain-mode patch cycles or image-version replacement, host by host, with no mid-session disconnects.

Scope — in

Golden-image build and capture; Azure Compute Gallery image definitions and versions; an image-build pipeline (manual-capture method plus the Azure VM Image Builder option); Bicep deployment of a host pool and session hosts from a gallery image version; FSLogix profile-container storage design and configuration on Azure Files / Azure NetApp Files; RBAC on the AVD objects; a scaling plan for cost

control; and the drain/patch update workflow.

Scope — out (covered elsewhere or by the customer)

- **End-user client access and connectivity** from Windows and macOS, workspace subscription, and the connection experience — see the companion **AVD Remote Access (Windows & macOS Clients)** solution.
- **Conditional Access / MFA policy design** for AVD sign-in — covered by the Remote Access solution and your Entra security baseline.
- **Which applications belong in the image**, application packaging/licensing, and **data-residency, records, and regulatory determinations** — these are customer business, compliance, and legal decisions; this guide provides the technical build, not the decision.
- **Production change approval** — disruptive steps (image cutover, host replacement, scaling changes) are staged behind your own change-management process.

Authentication and connections used

- **Azure RBAC** (Owner/Contributor to build; Desktop Virtualization roles to operate) on the subscription or resource group in scope.
- **Microsoft Entra ID** for the identity that session hosts and FSLogix storage join to (Entra join, Entra Domain Services, or AD DS — verify which your tenant uses).
- **Az PowerShell modules** (`Az.Accounts`, `Az.Compute`, `Az.DesktopVirtualization`, `Az.Storage`, `Az.Resources`) for the discovery, validation, reporting, and maintenance scripts.
- **Bicep / Azure Resource Manager** for the session-host and scaling-plan deployment template.

Feature availability, image SKUs, licensing, and any preview capabilities (for example Azure VM Image Builder regional availability or Entra-Kerberos for Azure Files) change over time — verify current availability in your tenant and region before you build.

2. Requirements — Azure Virtual Desktop: Deployment, Golden Image & FSLogix

The **category-level platform prerequisites** (an Entra tenant, admin roles, licensing baseline, network connectivity, change control, and backup/rollback posture) are covered once in `global-requirements-azure-solutions.md` — read that first. This document adds only the requirements **specific to building and operating AVD**: the image pipeline, IaC deployment, FSLogix storage, and scaling. Each item is structured **Description / Why required / How to verify / How to implement if missing**.

> *Naming convention (reused verbatim throughout): every object is named* `<Purpose> – <Scope> – <Stage>`*, for example* `AVD – Host Pool – Pilot` *and* `AVD – Session Host – Production`*. Azure resource types that forbid spaces/hyphens (notably the Compute Gallery) use the technical form noted at each object.*

R1. An AVD-eligible Windows license per user

• **Description:** Each user connecting to AVD holds an eligible license — Microsoft 365 E3/E5/A3/A5/F3/Business Premium, or Windows E3/E5, or a per-user access license for external users.

• **Why required:** Windows 11 Enterprise multi-session session hosts require an eligible license per connecting user; without it the deployment is out of compliance.

• **How to verify:** In the Microsoft 365 admin center → **Billing** → **Licenses**, confirm the in-scope users carry an eligible SKU. Cross-check the current eligible-license list in Microsoft's AVD documentation — *verify current availability, entitlements change.*

• **How to implement if missing:** Assign an eligible SKU to each in-scope user, or procure per-user access licensing for external users, before publishing the pilot.

R2. Azure Compute Gallery for image versioning

• **Description:** An Azure Compute Gallery (formerly Shared Image Gallery) with an **image definition** for the AVD image, into which numbered **image versions** are published. Logical name `AVD – Compute Gallery – Core`; because galleries forbid spaces and hyphens, the technical resource name is `avd_compute_gallery_core`, with image definition `avd-win11-m365-multisession`.

• **Why required:** The gallery is what makes the image *versioned, immutable, replicable, and roll-back-able*. Deploying from a Marketplace image directly gives you no version history and no rollback.

• **How to verify:** In the portal, open **Azure Compute Gallery** and confirm the gallery, the image definition (OS: Windows, generation: **Gen2**, security type matching your VMs), and at least one version exist. Or run `AVDDEP-Discovery.ps1`, which lists galleries, definitions, and versions.

• **How to implement if missing:** Create the gallery and an image definition (Section 3, Step 2). Set the image definition's OS state to **Generalized** and generation to **Gen2** to match modern AVD VM sizes.

R3. A build/source VM for the golden image

• **Description:** A temporary "build" VM — `AVD – Image Build VM – Staging` — from the correct base (Windows 11 Enterprise **multi-session** + Microsoft 365 Apps, Gen2) used to install and configure everything before capture.

• **Why required:** The golden image is produced by customizing a live VM and then generalizing/capturing it. A dedicated, disposable build VM keeps the process repeatable and keeps production hosts untouched during builds.

• **How to verify:** Confirm you can create a VM from the intended base image SKU in the target region (**Marketplace** → **Windows 11 Enterprise multi-session + Microsoft 365 Apps**), and that its size supports **Gen2 / Trusted Launch** if you use those.

• **How to implement if missing:** Deploy the build VM from the base image (Section 3, Step 3). Keep it in a dedicated `AVD – Resource Group – Imaging` resource group so build artifacts are isolated.

R4. FSLogix profile-container storage (Azure Files or Azure NetApp Files)

• **Description:** A file share for FSLogix profile containers — either an **Azure Files** premium (SMB) share or an **Azure NetApp Files** volume — sized for the user population, with an identity source (AD DS, Entra Domain Services, or Entra Kerberos) the session hosts can authenticate to. Logical name `AVD – Profile Share – Core`.

• **Why required:** On pooled multi-session hosts, profiles must live off-host so they roam. An undersized or high-latency share is the single most common cause of slow logons and profile corruption.

• **How to verify:** Confirm the share/volume exists, its tier (premium Azure Files or the correct ANF service level) and size match the plan (allow ~5–30 GB per profile depending on Outlook/Teams cache), and that session hosts can mount it with the chosen identity. `AVDDEP-Validate.ps1` checks that hosts have the FSLogix registry pointing at the share.

• **How to implement if missing:** Provision the share/volume, set NTFS and share-level permissions (users: modify; admins: full), join the storage to your identity source, and record the UNC path for the FSLogix configuration (Section 3, Step 6). *Entra Kerberos for Azure Files may be in preview in some clouds/regions — verify current availability.*

R5. Bicep / IaC tooling and deployment rights

• **Description:** The **Azure CLI** (or Az PowerShell) with **Bicep** installed, and RBAC rights to deploy resources into the target resource group (Contributor plus *Desktop Virtualization Contributor*).

• **Why required:** Session hosts are deployed from the provided `AVDDEP-SessionHosts.bicep` template. Deploying requires Bicep tooling and the rights to create VMs, NICs, and AVD host-pool resources.

• **How to verify:** Run `az bicep version` (or `Get-Command bicep`), and confirm your account holds Contributor + Desktop Virtualization Contributor on the target scope in **Subscriptions** → **Access control (IAM)**.

• **How to implement if missing:** Install Bicep (`az bicep install`), and have an Owner assign the required roles (least privilege — scope to the resource group, not the subscription, where possible).

R6. Host-pool registration path and networking

• **Description:** A virtual network/subnet in the target region for the session hosts, with line-of-sight to the FSLogix storage and to the identity source, and a **host-pool registration token** issued at deploy time so new hosts join the pool.

• **Why required:** Session-host VMs need a subnet, DNS that resolves your identity source, and a valid (unexpired) registration token to join the AVD host pool. An expired token is the classic cause of "VM deployed but not showing as a session host."

• **How to verify:** Confirm the subnet exists and resolves internal DNS; the Bicep deployment generates a fresh registration token, so verify the host pool's token expiry in **Host pool** → **Registration key** if you register hosts manually.

• **How to implement if missing:** Create/select the subnet, ensure DNS points at the identity source, and let the Bicep template mint a fresh token at deploy time (tokens are short-lived — generate at deploy, do not reuse old ones).

R7. Azure Monitor / Log Analytics workspace (recommended)

• **Description:** A Log Analytics workspace connected to the host pool and session hosts to surface connection, performance, and FSLogix diagnostics (AVD Insights).

• **Why required:** Validating "is the estate healthy and are logons fast?" depends on host and FSLogix telemetry. Without it, troubleshooting is blind.

• **How to verify:** In **Azure Virtual Desktop** → **Insights**, confirm a workspace is configured and hosts report data.

• **How to implement if missing:** Create a Log Analytics workspace and enable AVD Insights (diagnostic settings on the host pool + the AVD Insights workbook). This is recommended, not blocking, for the pilot.

Global Requirements — Azure Solutions

These requirements are common to **every** Azure Solutions solution in the standard program model. They are the baseline platform prerequisites that any Azure/Microsoft 365 engagement assumes. Each is structured **Description / Why / Verify / Implement-if-missing**. Solution-specific requirements (see `02-requirements.md`) are layered on top of these.

> *Naming convention: every created object is named `<Purpose> – <Scope> – <Stage>` and reused verbatim throughout the engagement.*

G1. A Microsoft 365 / Entra tenant

• **Description:** An active Microsoft 365 / Microsoft Entra tenant for the cloud(s) in scope (commercial and/or government).

• **Why:** All identity, licensing, and service configuration is anchored to a tenant; nothing can be configured without one.

• **Verify:** Sign in to the relevant admin center (`https://admin.microsoft.com`) and confirm the tenant resolves and is active under **Settings** → **Org settings**.

• **Implement-if-missing:** Provision the tenant through the appropriate channel (commercial: standard; government: via an authorized AOS-G reseller).

G2. Administrative roles (least privilege)

• **Description:** Appropriately scoped admin roles for the work — Global Administrator for tenant-wide changes, and service-specific roles (Exchange, SharePoint, Teams, Azure Owner/Contributor) for day-to-day.

• **Why:** Every configuration and migration action requires a role with sufficient privilege; least-privilege scoping limits blast radius and satisfies audit requirements.

• **Verify:** In Entra → **Roles & admins**, confirm the operating accounts hold the required roles. In Azure → **Subscriptions** → **Access control (IAM)**, confirm RBAC assignments.

• **Implement-if-missing:** Assign the minimum roles needed; prefer eligible (PIM) activation over standing access where available.

G3. Licensing baseline

• **Description:** Sufficient Microsoft 365 and/or Azure licensing/credit for all in-scope users and workloads.

• **Why:** Features and capacity (mailboxes, SharePoint storage, AVD, VMs) are gated by license SKUs and subscription quota.

• **Verify: Billing** → **Licenses** for M365; **Subscriptions** → **Usage + quotas** for Azure. Compare assigned/available against the in-scope object count.

• **Implement-if-missing:** Procure the correct SKUs and request quota increases sized to the workload (add headroom for shared/resource objects).

G4. Network connectivity to Microsoft 365 / Azure endpoints

• **Description:** Reliable network egress to the required Microsoft 365 and Azure service endpoints (and the sovereign `.us` endpoints for government clouds), with sufficient bandwidth.

• **Why:** All cloud administration and any data movement depends on connectivity to the correct endpoint set; government clouds use distinct endpoints from commercial.

• **Verify:** Confirm reachability to the relevant endpoints (commercial `*.microsoft.com` / government `*.microsoft.us`, `*.azure.us`, `*.office365.us`). Measure sustained bandwidth against the data volume.

• **Implement-if-missing:** Open firewall/proxy allowances for the correct endpoint set; scale bandwidth or add ExpressRoute (matching the cloud — Gov ExpressRoute for Gov regions).

G5. Change control & maintenance window

• **Description:** An approved change-management process with scheduled maintenance windows and a stakeholder communications plan.

• **Why:** Cloud changes can be outward-facing and disruptive; controlled windows and comms reduce business impact and satisfy governance/compliance.

• **Verify:** Confirm an approved change record and a scheduled window for each disruptive action.

• **Implement-if-missing:** Stand up a lightweight change/approval process; schedule windows; prepare user comms templates.

G6. Backup / rollback posture

• **Description:** A defined backup and rollback position for any change — config export, data backup, and a documented back-out procedure before any disruptive action.

• **Why:** Cloud changes are not always reversible by default; a tested rollback is the safety net for cutovers and bulk operations.

• **Verify:** Confirm pre-change config/state is captured and a written rollback exists for each disruptive change.

• **Implement-if-missing:** Export current configuration/state; document and (where feasible) rehearse the rollback before executing the change.

3. Implementation Guide — Azure Virtual Desktop: Deployment, Golden Image & FSLogix

Follow these steps in order. Every object you create is named with the convention `<Purpose> – <Scope> – <Stage>` and reused verbatim in later steps, in **Testing**, and in the scripts. Portal paths are current as of authoring; Microsoft renames blades occasionally — *verify the exact label in your portal*. **Pilot first**: build one image, one pilot host pool, prove it, then expand.

> *Before you start, run `AVDDEP-Discovery.ps1` to capture what already exists (galleries, host pools, session hosts, storage, scaling plans). This is your baseline and your rollback reference (global requirement G6).*

Step 1 — Create the resource groups and the Compute Gallery

1. In the Azure portal, go to **Resource groups** → **+ Create** and create two groups in your target region:

• `AVD – Resource Group – Core` — holds the host pool, workspace, application group, and scaling plan.

• `AVD – Resource Group – Imaging` — holds the build VM and the Compute Gallery, so build artifacts stay isolated.

1. Go to **Azure Compute Gallery** → **+ Create**, in `AVD – Resource Group – Imaging`. Give it the technical name `avd_compute_gallery_core` (galleries forbid spaces and hyphens; this is the logical `AVD – Compute Gallery – Core`). Create.

2. Open the gallery → **Definitions** → **+ Add / Create a VM image definition**:

• **Name:** `avd-win11-m365-multisession`

• **OS type:** Windows · **OS state:** Generalized · **VM generation: Gen2**

• Publisher/Offer/SKU: your own identifiers (e.g. `contoso / avd / win11-multisession`).

• Create. You now have a definition ready to receive versions.

Step 2 — Deploy the build VM

1. Go to **Virtual machines** → **+ Create**. Under **Image**, choose **See all images** → **Marketplace** and select **Windows 11 Enterprise multi-session + Microsoft 365 Apps** (Gen2). *Verify the exact SKU name in Marketplace — Microsoft revises image names.*

2. Name it `AVD – Image Build VM – Staging`, place it in `AVD – Resource Group – Imaging`, size it modestly (e.g. a D4s-class VM — this is a build box, not a session host), and place it on your build subnet.

3. Leave it **not** joined to your production identity during build (a workgroup build VM is cleanest for generalizing). Create and connect over RDP/Bastion.

Step 3 — Build the golden image (customize the source)

On `AVD – Image Build VM – Staging`, apply your standard build. Keep a written checklist so every future rebuild is identical:

1. Run **Windows Update** fully; reboot until clean.
2. Install the **FSLogix agent** (latest) — configuration comes later via policy/registry, not baked-in secrets.
3. Install line-of-business apps and runtimes that belong on every host.
4. Apply the **Microsoft AVD image optimizations** (the Virtual Desktop Optimization Tool / recommended settings) to trim background tasks.
5. Do **not** run Sysprep manually yet if you will capture through the portal (the capture flow generalizes for you). Remove any per-user or secret material.

*Everything disruptive here is on a disposable build VM, so there is no production impact. This is the only place app contents are decided — *which* apps belong in the image is a customer decision (Overview, "Scope — out").*

Step 4 — Capture the image as a gallery version

1. On the build VM's blade, click **Capture** (top toolbar).
2. Choose **Yes, share it to a gallery as a VM image version.**
3. Target gallery `avd_compute_gallery_core`, image definition `avd-win11-m365-multisession`.
4. **Version number:** `1.0.0` (use semantic versions; a monthly patch becomes `1.1.0`, a rebuild-from-scratch `2.0.0`).
5. Set **replica count** and **target regions** to match where you will deploy hosts (replicate to each region you run host pools in).
6. Tick **Automatically delete this virtual machine after capture** so the build VM is cleaned up.
7. Create. When the version shows **Succeeded**, you have an immutable, versioned golden image. Confirm with `AVDDEP-Report.ps1` (image inventory).

Step 5 — Prepare FSLogix profile storage

1. Provision the profile share `AVD – Profile Share – Core` — either an **Azure Files** premium file share or an **Azure NetApp Files** volume — in the host-pool region (co-locate with hosts for low latency).
2. Join the storage to your identity source (AD DS, Entra Domain Services, or Entra Kerberos for Azure Files — *verify current availability in your cloud/region*).

3. Set permissions: **Users** → **Modify** on the share (so each user can create their own container), **Admins** → **Full**. Remove broad inherited rights.

4. Record the UNC path (e.g. `\\avdprofilescore.file.core.windows.net\profiles`). You will point FSLogix at it in Step 8.

Step 6 — Deploy the pilot host pool and session hosts with Bicep

1. Open the provided `AVDDEP-SessionHosts.bicep` and set its parameters:

• `hostPoolName`: `AVD – Host Pool – Pilot`

• `hostPoolType`: `Pooled`, `loadBalancerType`: `BreadthFirst`

• `galleryImageId`: the resource ID of version `1.0.0` of `avd-win11-m365-multisession`

• `sessionHostCount`: **2** (pilot), `vmSize`: your standard multi-session size

• `subnetId`, `resourceGroup`: `AVD – Resource Group – Core`, `location`: your region

1. **Preview first (never apply blind):**

```
az deployment group what-if \
  --resource-group "AVD – Resource Group – Core" \
  --template-file AVDDEP-SessionHosts.bicep \
  --parameters @avddep.pilot.parameters.json
```

Review the what-if output — it must show the host pool, 2 VMs/NICs, and the scaling-plan association being **created**, nothing being deleted.

1. Deploy (swap `what-if` for `create`). The template mints a **fresh host-pool registration token** at deploy time (requirement R6) so the new hosts join `AVD – Host Pool – Pilot` automatically.

2. Confirm the two hosts appear under **Host pool** → **Session hosts** as **Available**. Run `AVDDEP-Validate.ps1` to confirm agent health and image provenance.

*The host pool, workspace, and application-group *plumbing* is created here; **publishing to end users and validating the Windows/macOS client** is the companion *AVD Remote Access* solution — do not duplicate that work here.*

Step 7 — Assign RBAC on the AVD objects

1. On `AVD – Host Pool – Pilot` → **Access control (IAM)** → **+ Add role assignment**, assign **Desktop Virtualization Host Pool Contributor** to the operators group `AVD – Operators – Group`.

2. On `AVD – Resource Group – Core`, assign **Desktop Virtualization Contributor** to the same group for day-to-day operation, and keep **Owner** limited to the platform team.

3. Prefer eligible (PIM) activation over standing access where your licensing allows (global requirement G2). Confirm with `AVDDEP-Validate.ps1` (RBAC section).

Step 8 — Configure FSLogix on the session hosts

1. FSLogix is installed in the image (Step 3); now point it at the share. Apply the FSLogix settings by **Group Policy** or the registry values under `HKLM\SOFTWARE\FSLogix\Profiles`:

• `Enabled = 1`

• `VHDLocations = \\avdprofilescore.file.core.windows.net\profiles` (the Step 5 UNC path)

• `DeleteLocalProfileWhenVHDShouldApply = 1`, `FlipFlopProfileDirectoryName = 1`

1. Sign a pilot user in twice on different hosts and confirm a `Profile_<username>.vhdx` appears on the share and the profile roams. `AVDDEP-Validate.ps1` reports the FSLogix registry state per host.

Step 9 — Create and assign the autoscale scaling plan

1. Go to **Azure Virtual Desktop** → **Scaling plans** → **+ Create**. Name it `AVD – Scaling Plan – Pilot`, region matching the host pool, **host pool type: Pooled**.

2. Add a schedule (e.g. `Weekdays`): ramp-up before business hours, peak during the day, ramp-down and off-peak after hours, with a **capacity threshold** (e.g. 75–80%) and **minimum percent of hosts** during ramp-down.

3. Under **Host pool assignments**, tick `AVD – Host Pool – Pilot` and **Enable autoscale**.

4. Grant the AVD service principal the **Power On/Off** rights it needs (the portal prompts for the required custom role assignment). Confirm the plan and schedule with `AVDDEP-Report.ps1` (scaling section).

Step 10 — Establish the drain-and-patch workflow

1. When a new image version (e.g. `1.1.0`) is ready, you patch by **replacing hosts**, not editing live ones: put existing hosts into **drain mode**, let sessions finish, then deploy replacement hosts from the new version and remove the old.

2. Preview the drain safely with `AVDDEP-Maintenance.ps1` — it **defaults to** `-WhatIf`, listing which hosts *would* be drained and their active session counts. Only with an explicit `-Apply` does it set drain mode.

3. Re-run `AVDDEP-SessionHosts.bicep` with `galleryImageId` pointed at `1.1.0` to add fresh hosts, confirm health, then remove the drained old hosts. This is a **change-controlled** action (global requirements G5/G6) — do it in a window with a rollback (redeploy the prior image version) ready.

Step 11 — Validate the pilot, then expand

1. Run the full **Testing** guide (`05-testing.md`) against `AVD – Host Pool – Pilot`. All gates must pass before expanding.

2. To expand, reuse the same artifacts: bump `sessionHostCount` and deploy a `AVD – Host Pool – Production` from the **same** image version and Bicep template, attach a `AVD – Scaling Plan – Production`, and assign RBAC. No hand-building — every wave is the same reviewable deployment.

Order of operations (summary)

Resource groups + gallery → build VM → customize → capture version `1.0.0` → FSLogix storage → **what-if** then deploy pilot hosts (Bicep) → RBAC → FSLogix config → scaling plan → drain/patch workflow → validate → expand to production from the same image and template.

5. Testing & Validation — Azure Virtual Desktop: Deployment, Golden Image & FSLogix

Run every check against the pilot (`AVD – Host Pool – Pilot`) before expanding to production. Each test has an explicit **pass/fail gate**. The scripts referenced here are in the `scripts/` folder and are read-only except `AVDDEP-Maintenance.ps1`, which defaults to a safe `-WhatIf` preview.

> *Run order:* `AVDDEP-Discovery.ps1` → `AVDDEP-Validate.ps1` → `AVDDEP-Report.ps1`. *Use* `AVDDEP-Maintenance.ps1` *(preview mode) only to rehearse the drain/patch gate.*

Test 1 — Golden image is versioned in the gallery

- **How:** Run `AVDDEP-Discovery.ps1` (or `AVDDEP-Report.ps1`). Confirm `avd_compute_gallery_core` contains the definition `avd-win11-m365-multisession` with version `1.0.0` in state **Succeeded**, replicated to each target region.
- **Pass:** The version exists, is Succeeded, and is Gen2/Generalized.
- **Fail:** No version, a Failed replication, or a generation mismatch → revisit Section 3, Step 4.

Test 2 — Session hosts were built from the intended image version

- **How:** Run `AVDDEP-Validate.ps1`. It reports each session host's source image reference and flags any host whose image does **not** match the gallery version the host pool was deployed from.
- **Pass:** All hosts in `AVD – Host Pool – Pilot` trace to gallery version `1.0.0`.
- **Fail:** A host built from a Marketplace image or an older version → it is a snowflake; redeploy it from Bicep (Step 6).

Test 3 — Session hosts are healthy and registered

- **How:** In **Host pool** → **Session hosts**, confirm both pilot hosts show **Available** with a healthy AVD agent; `AVDDEP-Validate.ps1` reports agent/status per host.
- **Pass:** All pilot hosts **Available**, no **Unavailable**/**Upgrading**/**NeedsAssistance**.
- **Fail:** A host stuck not-registered usually means an expired registration token (requirement R6) → redeploy so a fresh token is minted; do not hand-edit.

Test 4 — FSLogix profiles roam

- **How:** Sign a pilot user into two different hosts in turn. Confirm a `Profile_<user>.vhdx` is created on `AVD – Profile Share – Core` and that desktop/settings/Outlook cache follow the user. `AVDDEP-Validate.ps1` confirms the FSLogix registry on each host points at the correct UNC path with `Enabled=1`.
- **Pass:** One container per user on the share; profile identical across hosts; logon completes in a reasonable time.

• **Fail:** A fresh temp profile each logon, or no VHDX → check share permissions (users: Modify) and the `VHDLocations` value (Section 3, Step 8).

Test 5 — RBAC is least-privilege and correct

• **How:** `AVDDEP-Validate.ps1` lists role assignments on the host pool and resource group. Confirm operators hold **Desktop Virtualization** roles, not broad **Owner/Contributor** at subscription scope.

• **Pass:** Operators have Desktop Virtualization Contributor / Host Pool Contributor scoped to the RG/host pool; Owner is limited to the platform team.

• **Fail:** Broad standing Owner on operators → tighten per Section 3, Step 7 (prefer PIM).

Test 6 — Scaling plan controls capacity

• **How:** `AVDDEP-Report.ps1` prints the scaling plan, its schedules, and the host-pool assignment. Confirm `AVD – Scaling Plan – Pilot` is **enabled** and assigned to `AVD – Host Pool – Pilot`. Observe that off-peak drives idle hosts to deallocated.

• **Pass:** Plan enabled and assigned; ramp-up/ramp-down/off-peak schedule present; idle hosts deallocate off-peak.

• **Fail:** Plan not assigned, or hosts never power down → confirm the AVD service principal has Power On/Off rights (Section 3, Step 9).

Test 7 — IaC deployment is reproducible (what-if is clean)

• **How:** Re-run the Section 3, Step 6 `az deployment group what-if` against the pilot parameters.

• **Pass:** What-if reports **no changes** (the deployed state matches the template) — proving the estate is fully described by code.

• **Fail:** What-if wants to change/recreate resources you did not change → a manual drift crept in; reconcile the template to reality before expanding.

Test 8 — Drain/patch workflow is safe (rehearsal)

• **How:** Run `AVDDEP-Maintenance.ps1` in its **default** `-WhatIf` mode. Confirm it lists which hosts *would* be drained and their active session counts **without changing anything**.

• **Pass:** Preview lists targets and session counts; no drain mode is actually set.

• **Fail:** Any state change in preview mode, or the script errors → do not run `-Apply`; resolve first.

Exit criteria (all must be green before expanding)

Image version Succeeded (T1) · hosts trace to that version (T2) · hosts Available (T3) · FSLogix roams (T4) · RBAC least-privilege (T5) · scaling plan enabled and cost-effective (T6) · what-if clean/reproducible (T7) · drain preview safe (T8). Only then deploy `AVD – Host Pool – Production` from the same image and template.

6. Troubleshooting — Azure Virtual Desktop: Deployment, Golden Image & FSLogix

Common failures on the build/deploy/FSLogix/scaling side, as **symptom** → **cause** → **fix**. Run `AVDDEP-Discovery.ps1` and `AVDDEP-Validate.ps1` first — most issues show up there. (Client-connection problems from Windows/macOS belong to the companion *AVD Remote Access* solution.)

Image build & gallery

Symptom — image version stuck in *Creating* or *Failed* replication.

- **Cause:** Region replication in progress, quota shortfall in a target region, or a generalize/capture that did not complete cleanly.
- **Fix:** Check the version's replication status per region; ensure compute quota exists in each target region; if the capture failed, redeploy the build VM and re-capture. Confirm the build VM was **generalized** (OS state *Generalized*) before capture.

Symptom — new session hosts won't build from the gallery image ("image not found" / generation mismatch).

- **Cause:** The image definition generation (Gen1/Gen2) or security type does not match the VM size, or the `galleryImageId` points at a definition instead of a specific version.
- **Fix:** Ensure the definition is **Gen2** and the VM size supports it (and Trusted Launch if used). Point `galleryImageId` at a full **version** resource ID (`.../versions/1.0.0`), not the definition.

Symptom — every host looks slightly different despite "using the image."

- **Cause:** Some hosts were created from a Marketplace image or an older version, i.e. snowflakes.
- **Fix:** `AVDDEP-Validate.ps1` (Test 2) flags hosts whose image does not trace to the expected version — redeploy those from `AVDDEP-SessionHosts.bicep` pinned to the correct version.

Session-host deployment (Bicep)

Symptom — VMs deploy but never appear as session hosts in the host pool.

- **Cause:** Expired or reused host-pool **registration token** (the single most common cause).
- **Fix:** Registration tokens are short-lived. Let the Bicep template mint a fresh token at deploy time (do not reuse an old one). Redeploy so a new token is issued, then confirm hosts appear as **Available**.

Symptom — `what-if` wants to delete or recreate resources you did not change.

- **Cause:** Manual drift — someone edited the estate in the portal after the last deploy.
- **Fix:** Reconcile the template to reality (import the manual change into the Bicep or revert the manual change) until `what-if` reports **no changes** (Testing T7). Never expand to production on a dirty what-if.

Symptom — deployment fails on the admin password parameter.

- **Cause:** Password complexity not met, or the secure parameter was left empty.

• **Fix:** Supply a compliant password via a **secure parameter or Key Vault reference** — never hard-code it. Meet Windows complexity rules (length + character classes).

FSLogix / profiles

Symptom — users get a fresh temporary profile every logon; no VHDX on the share.

• **Cause:** FSLogix not enabled, wrong `VHDLocations`, or the user lacks Modify rights on the share.

• **Fix:** Confirm `HKLM\SOFTWARE\FSLogix\Profiles` has `Enabled=1` and the correct UNC in `VHDLocations` (`AVDDEP-Validate.ps1`, Test 4). Set share permissions so **Users = Modify**. Verify the storage is joined to the identity source the hosts authenticate to.

Symptom — logons are very slow.

• **Cause:** Undersized/high-latency profile storage, cross-region profile I/O, or antivirus scanning the VHDX.

• **Fix:** Use **premium** Azure Files or the correct Azure NetApp Files service level, **co-located** with the hosts. Exclude the FSLogix VHDX path from AV real-time scanning per your vendor's AVD guidance.

Symptom — "profile already in use" / users locked out of their profile.

• **Cause:** A prior session did not release the container (host crash), or the same user is active on two hosts.

• **Fix:** Confirm breadth-first load balancing and single-session-per-user behavior; clear the stale lock on the container; ensure `DeleteLocalProfileWhenVHDShouldApply=1` so local remnants don't conflict.

RBAC

Symptom — operators can't manage the host pool, or hold far more rights than they should.

• **Cause:** Missing Desktop Virtualization role, or broad standing Owner/Contributor at subscription scope.

• **Fix:** Assign **Desktop Virtualization Host Pool Contributor** / **Desktop Virtualization Contributor** scoped to the host pool/RG; remove broad Owner from operators and prefer **PIM** eligibility (`AVDDEP-Validate.ps1`, Test 5).

Scaling & cost

Symptom — idle hosts never power down; the bill keeps climbing.

• **Cause:** No scaling plan assigned, plan disabled, or the AVD service principal lacks Power On/Off rights.

• **Fix:** Confirm `AVD – Scaling Plan – Pilot` is **enabled and assigned** to the host pool (`AVDDEP-Report.ps1`), and that the required custom role for power actions is granted (the portal prompts for it during plan creation).

Symptom — users get disconnected during the day.

• **Cause:** Ramp-down schedule set too aggressively / `rampDownForceLogoffUsers` enabled during business hours.

• **Fix:** Review the schedule times and thresholds in the scaling plan; keep force-logoff off for interactive audiences and set a sensible `rampDownWaitTimeMinutes`.

Drain / patch

Symptom — updating a host disconnected an active user mid-session.

• **Cause:** A host was patched or replaced without draining first.

• **Fix:** Always **drain then replace**. Preview with `AVDDEP-Maintenance.ps1` (default `-WhatIf`) to see active session counts, drain with `-Apply` in a maintenance window, let sessions finish, then deploy replacement hosts from the new image version. Rollback = `-Undrain -Apply` or redeploy the prior image version.

7. Best Practices — Azure Virtual Desktop: Deployment, Golden Image & FSLogix

Keep the estate clean, repeatable, and cheap. These practices keep hosts identical, profiles fast, and the bill predictable.

Golden image discipline

• **One source of truth.** Every session host comes from a **gallery image version** — never a hand-patched live host. If you edit a host directly, you have created a snowflake; rebuild it from the image instead.

• **Semantic versioning with intent.** Monthly Windows/app patch → minor bump (`1.0.0` → `1.1.0`); a from-scratch rebuild or major OS change → major bump (`2.0.0`). The version number should tell you what changed.

• **Keep a written build checklist** (Section 3, Step 3) so any engineer can reproduce the image. Better still, graduate the manual capture to **Azure VM Image Builder** so the definition rebuilds on a schedule — *verify VM Image Builder availability in your region/cloud.*

• **Replicate versions** to every region you run hosts in, and **retain the previous version** so you can roll a host pool back if a new image regresses.

FSLogix and profile performance

• **Co-locate** the profile share with the session hosts (same region) — cross-region profile I/O is the top cause of slow logons.

• **Size for the real profile**, including Outlook cached mode and Teams cache; premium Azure Files or the right Azure NetApp Files service level, not standard storage.

• **Least-privilege share permissions** (users: Modify, admins: Full) so users create only their own container.

• **Exclude** the FSLogix VHDX path from on-host antivirus real-time scanning to avoid logon stalls (follow your AV vendor's AVD guidance).

Deployment and change control

• **What-if before every apply.** The Bicep `what-if` (Testing T7) is your drift detector and your change record — a clean what-if means the estate equals the code.

• **Fresh registration tokens at deploy time.** Never reuse an old token; let the template mint one. Expired tokens are the classic "VM deployed but not a session host" failure.

• **Patch by replacement, not by hand.** Use the drain-and-patch workflow (Section 3, Step 10): drain old hosts, deploy new ones from the new image version, remove the old — no mid-session disconnects, and a clean rollback (redeploy the prior version).

Cost control

• **Always attach a scaling plan.** Idle multi-session hosts running overnight are the biggest avoidable cost. Tune ramp-down aggressively for non-24×7 audiences.

• **Right-size the VM SKU** to real session density; measure with AVD Insights before scaling out.

• **Deallocate the imaging build VM** (auto-delete after capture) so build boxes never linger.

Security and governance

• **Least-privilege RBAC** with the Desktop Virtualization roles; prefer **PIM** eligible activation over standing Owner.

• **Defer the business decisions** — which apps belong in the image, data-residency and records requirements, and production change approval are customer compliance/legal/ops calls, not technical defaults (Overview, "Scope — out").

• **Enable AVD Insights / Log Analytics** so health, logon time, and FSLogix errors are observable, not guessed.

Where to go next (adjacent catalog areas)

• **Azure Virtual Desktop Remote Access (Windows & macOS Clients)** — the companion solution for publishing this estate to end users and validating client connectivity and Conditional Access. Deploy this build solution first, then that one.

• **Azure Security Baseline & Governance** — Azure Policy, RBAC, and guardrails to wrap the subscription your AVD estate lives in.

• **Windows 365 Cloud PC Deployment** — for personal, always-on Cloud PCs where per-user simplicity beats pooled multi-session economics; useful to compare against pooled AVD for parts of your population.

• **Azure Kubernetes Service Secure Baseline** — if the same team runs containerized workloads alongside AVD and wants the same IaC-and-baseline rigor.

8. Deliverables Summary — Azure Virtual Desktop: Deployment, Golden Image & FSLogix

Everything in this solution, at a glance. This is the *build and operate* side of AVD (imaging, IaC deployment, FSLogix storage, RBAC, scaling); end-user client access from Windows/macOS is the companion **AVD Remote Access** solution.

Documentation (source/)

Document	Purpose
`executive-summary.md`	What the solution is, the scope boundary vs. AVD Remote Access, what you get, and the effort.
`01-solution-overview.md`	How golden images, Compute Gallery versioning, image pipelines, IaC deployment, FSLogix, RBAC, drain/patch, and scaling work; the Assess → Configure → Pilot → Validate → Expand model; in/out of scope.
`02-requirements.md`	Build-specific prerequisites (license, Compute Gallery, build VM, FSLogix storage, Bicep tooling, networking, monitoring) with verify / implement-if-missing. References the shared `global-requirements-azure-solutions.md`.
`03-implementation.md`	Ordered, click-by-click build: resource groups + gallery, build VM, customize, capture version `1.0.0`, FSLogix storage, what-if-then-deploy pilot hosts by Bicep, RBAC, FSLogix config, scaling plan, drain/patch, then expand. Every object named `<Purpose> – <Scope> – <Stage>`.
`05-testing.md`	Eight pass/fail validation gates (image versioned, hosts trace to version, hosts healthy, FSLogix roams, RBAC least-privilege, scaling effective, what-if clean, drain safe) with exit criteria.
`06-troubleshooting.md`	Symptom → cause → fix for image/gallery, Bicep deployment, FSLogix, RBAC, scaling, and drain/patch failures.
`07-best-practices.md`	Golden-image discipline, FSLogix performance, change control, cost control, security, and where to go next in the catalog.
`08-deliverables-summary.md`	This inventory.
`global-requirements-azure-solutions.md`	Shared Azure category prerequisites (tenant, roles, licensing, connectivity, change control, rollback).

Automation (scripts/)

Script	Purpose
`AVDDEP-Discovery.ps1`	**Read-only.** Inventories the Azure context, Compute Galleries/definitions/versions, host pools and session hosts (with image provenance), profile storage, and scaling plans — your pre-change baseline.
`AVDDEP-Validate.ps1`	**Read-only.** Runs the Testing gates: image provenance, host availability, FSLogix registry, least-privilege RBAC, and scaling-plan assignment, printing PASS/FAIL per check.
`AVDDEP-Report.ps1`	**Read-only.** Produces an inventory/cost-posture report — per-host size, drain state and active sessions, image versions in use, and scaling schedules (optional CSV export).
`AVDDEP-SessionHosts.bicep`	**Infrastructure-as-code.** Deploys the host pool, workspace + Desktop app group, N session hosts from a specified gallery image version, and a scaling plan — with a fresh registration token minted at deploy time. Always run `what-if` first.
`AVDDEP-Maintenance.ps1`	**Gated change (defaults to `-WhatIf` preview).** Drain-and-patch workflow: previews which hosts would be drained and their session counts; only `-Apply` sets/lifts drain mode, ahead of image-version replacement.

Package

Delivered as a **Complete Solution Package (PDF + Word + scripts)** — the eight guides above compiled to a Word document and a PDF, plus all five automation artifacts (three PowerShell scripts, one Bicep template, one gated PowerShell maintenance script) in a single ZIP — yours to keep and reuse on every AVD rollout.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

AVDDEP-Discovery.ps1

Purpose: Captures a read-only baseline of everything on the deployment/image side of AVD before you build or change anything: the signed-in Azure context, Azure Compute Galleries with their image definitions and versions, host pools with their session hosts and source image, any profile storage accounts, and scaling plans. Use it as your pre-change baseline (rollback reference). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\AVDDEP-Discovery.ps1 `-ResourceGroupName 'AVD - Resource Group - Core'` -ImagingResourceGroupName 'AVD - Resource Group - Imaging' Requires Az.Accounts, Az.Compute, Az.DesktopVirtualization, Az.Storage. Run Connect-AzAccount first (or the script prompts). Read-only. EXPECTED RESULTS Azure context; galleries/definitions/versions; host pools with session-host counts and image provenance; profile storage accounts; scaling plans. Nothing is modified. .NOTES Azure Solutions — Azure Virtual Desktop (deployment/image/FSLogix/scaling). Read-only. > [CmdletBinding()] param([string]$ResourceGroupName = 'AVD - Resource Group - Core', [string]$ImagingResourceGroupName = 'AVD - Resource Group - Imaging')

How to run: powershell -ExecutionPolicy Bypass -File .\AVDDEP-Discovery.ps1 `-ResourceGroupName 'AVD - Resource Group - Core'` -ImagingResourceGroupName 'AVD - Resource Group - Imaging' Requires Az.Accounts, Az.Compute, Az.DesktopVirtualization, Az.Storage. Run Connect-AzAccount first (or the script prompts). Read-only. EXPECTED RESULTS Azure context; galleries/definitions/versions; host pools with session-host counts and image provenance; profile storage accounts; scaling plans. Nothing is modified. .NOTES Azure Solutions — Azure Virtual Desktop (deployment/image/FSLogix/scaling). Read-only. > [CmdletBinding()] param([string]$ResourceGroupName = 'AVD - Resource Group - Core', [string]$ImagingResourceGroupName = 'AVD - Resource Group - Imaging')

AVDDEP-Maintenance.ps1

Purpose: Supports the patch-by-replacement workflow (03-implementation Step 10). By DEFAULT it runs in preview (-WhatIf) mode: it lists which hosts WOULD be drained and their active session counts, and changes NOTHING. Only with the explicit -Apply switch does it set AllowNewSession=$false (drain mode) so existing sessions finish while no new sessions land — after which you deploy replacement hosts from the new gallery image version and remove the drained hosts. HOW TO RUN 1) PREVIEW (default, read-only) — always do this first: powershell -ExecutionPolicy Bypass -File .\AVDDEP-Maintenance.ps1 `-ResourceGroupName 'AVD - Resource Group - Core' -HostPoolName 'AVD - Host Pool - Pilot' 2) APPLY drain (change-controlled; run in a maintenance window with rollback ready): powershell -ExecutionPolicy Bypass -File .\AVDDEP-Maintenance.ps1` -ResourceGroupName 'AVD - Resource Group - Core' -HostPoolName 'AVD - Host Pool - Pilot' ` -HostNames 'avdsh-pilot-0','avdsh-pilot-1' -Apply To lift drain again, add -Undrain (with -Apply). Requires Az.Accounts, Az.DesktopVirtualization. Rollback: re-run with -Undrain -Apply, or redeploy the prior gallery image version via AVDDEP-SessionHosts.bicep. EXPECTED RESULTS Preview: a table of hosts, current AllowNewSession state, and active session counts — no change. Apply: drain mode set (or lifted with -Undrain) on the named hosts, with confirmation output. .NOTES Azure Solutions — Azure Virtual Desktop (deployment/image/FSLogix/scaling). Change action GATED behind -Apply; defaults to a safe read-only preview. > [CmdletBinding(SupportsShouldProcess, ConfirmImpact =

'High')] param([Parameter(Mandatory)][string]$ResourceGroupName, [Parameter(Mandatory)][string]$HostPoolName, [string[]]$HostNames = @(), # empty in preview = show all hosts [switch]$Undrain, # lift drain instead of setting it [switch]$Apply # required to make any change; otherwise preview only)

How to run: 1) PREVIEW (default, read-only) — always do this first: powershell -ExecutionPolicy Bypass -File .\AVDDEP-Maintenance.ps1 `-ResourceGroupName 'AVD - Resource Group - Core' -HostPoolName 'AVD - Host Pool - Pilot'` 2) APPLY drain (change-controlled; run in a maintenance window with rollback ready): `powershell -ExecutionPolicy Bypass -File .\AVDDEP-Maintenance.ps1` -ResourceGroupName 'AVD - Resource Group - Core' -HostPoolName 'AVD - Host Pool - Pilot' ` -HostNames 'avdsh-pilot-0','avdsh-pilot-1' -Apply To lift drain again, add -Undrain (with -Apply). Requires Az.Accounts, Az.DesktopVirtualization. Rollback: re-run with -Undrain -Apply, or redeploy the prior gallery image version via AVDDEP-SessionHosts.bicep. EXPECTED RESULTS Preview: a table of hosts, current AllowNewSession state, and active session counts — no change. Apply: drain mode set (or lifted with -Undrain) on the named hosts, with confirmation output. .NOTES Azure Solutions — Azure Virtual Desktop (deployment/image/FSLogix/scaling). Change action GATED behind -Apply; defaults to a safe read-only preview. > [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'High')] param([Parameter(Mandatory)][string]$ResourceGroupName, [Parameter(Mandatory)][string]$HostPoolName, [string[]]$HostNames = @(), # empty in preview = show all hosts [switch]$Undrain, # lift drain instead of setting it [switch]$Apply # required to make any change; otherwise preview only)

AVDDEP-Report.ps1

Purpose: Generates a concise report for review or handover: the gallery image versions in use, session hosts with their VM size, drain/allow-new-session state and active session count, and the scaling plan schedules that govern cost. Optionally exports to CSV. Read-only. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\AVDDEP-Report.ps1 `-ResourceGroupName 'AVD - Resource Group - Core'` -HostPoolName 'AVD - Host Pool - Pilot' ` -CsvPath .\AVDDEP-report.csv # optional Requires Az.Accounts, Az.Compute, Az.DesktopVirtualization. Read-only. EXPECTED RESULTS A per-host table (size, drain state, sessions), the gallery versions referenced, and the scaling schedule summary. If -CsvPath is given, the host table is also written to CSV. .NOTES Azure Solutions — Azure Virtual Desktop (deployment/image/FSLogix/scaling). Read-only. > [CmdletBinding()] param([Parameter(Mandatory)][string]$ResourceGroupName, [Parameter(Mandatory)][string]$HostPoolName, [string]$CsvPath = '')

How to run: powershell -ExecutionPolicy Bypass -File .\AVDDEP-Report.ps1 `-ResourceGroupName 'AVD - Resource Group - Core'` -HostPoolName 'AVD - Host Pool - Pilot' ` -CsvPath .\AVDDEP-report.csv # optional Requires Az.Accounts, Az.Compute, Az.DesktopVirtualization. Read-only. EXPECTED RESULTS A per-host table (size, drain state, sessions), the gallery versions referenced, and the scaling schedule summary. If -CsvPath is given, the host table is also written to CSV. .NOTES Azure Solutions — Azure Virtual Desktop (deployment/image/FSLogix/scaling). Read-only. > [CmdletBinding()] param([Parameter(Mandatory)][string]$ResourceGroupName, [Parameter(Mandatory)][string]$HostPoolName, [string]$CsvPath = '')

AVDDEP-SessionHosts.bicep

Purpose: // Creates (or updates) a pooled host pool, a workspace + Desktop application // group, N session-host VMs built from a specified gallery image version, and // a scaling plan bound to the host pool.

A fresh host-pool registration token // is minted at deploy time so new hosts join automatically. // // HOW TO RUN — ALWAYS what-if FIRST, then create: // az deployment group what-if \ // --resource-group "AVD - Resource Group - Core" \ // --template-file AVDDEP-SessionHosts.bicep \ // --parameters @avddep.pilot.parameters.json // # review output — must CREATE the pool/VMs/plan, DELETE nothing — then: // az deployment group create ... (same args) // // EXPECTED RESULTS // A host pool ('AVD - Host Pool - Pilot' by default), workspace, Desktop app // group, `sessionHostCount` VMs from image version `galleryImageId`, and an // enabled scaling plan. Pilot with sessionHostCount=2; expand by raising it. // // NOTES // White-label. Pilot-first and plan/what-if-first by design. Do not embed // secrets: pass adminPassword via a secure parameter or Key Vault reference, // never hard-coded. Verify VM SKU/image availability in your region. // ==========

AVDDEP-Validate.ps1

Purpose: Read-only pass/fail checks that map to the Testing guide (05-testing.md): every session host in the target host pool traces to the expected gallery image version; hosts are Available with a healthy agent; FSLogix registry points at the profile share; RBAC on the host pool uses the Desktop Virtualization roles; and a scaling plan is assigned. Prints PASS/FAIL per check. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\AVDDEP-Validate.ps1 `-ResourceGroupName 'AVD - Resource Group - Core'` -HostPoolName 'AVD - Host Pool - Pilot' `-ExpectedImageVersion '1.0.0'` -ExpectedProfileUnc '\\avdprofilescore.file.core.windows.net\profiles' Requires Az.Accounts, Az.DesktopVirtualization, Az.Resources. FSLogix registry checks that include remote hosts require WinRM to the session hosts; otherwise run the FSLogix section locally on a host. Read-only. EXPECTED RESULTS A PASS/FAIL line per gate. Any FAIL points back to the matching step in 03-implementation.md. .NOTES Azure Solutions — Azure Virtual Desktop (deployment/image/FSLogix/scaling). Read-only. > [CmdletBinding()] param([Parameter(Mandatory)][string]$ResourceGroupName, [Parameter(Mandatory)][string]$HostPoolName, [string]$ExpectedImageVersion = '1.0.0', [string]$ExpectedProfileUnc = '', [string[]]$SessionHostComputerNames = @())

How to run: powershell -ExecutionPolicy Bypass -File .\AVDDEP-Validate.ps1 `-ResourceGroupName 'AVD - Resource Group - Core'` -HostPoolName 'AVD - Host Pool - Pilot' `-ExpectedImageVersion '1.0.0'` -ExpectedProfileUnc '\\avdprofilescore.file.core.windows.net\profiles' Requires Az.Accounts, Az.DesktopVirtualization, Az.Resources. FSLogix registry checks that include remote hosts require WinRM to the session hosts; otherwise run the FSLogix section locally on a host. Read-only. EXPECTED RESULTS A PASS/FAIL line per gate. Any FAIL points back to the matching step in 03-implementation.md. .NOTES Azure Solutions — Azure Virtual Desktop (deployment/image/FSLogix/scaling). Read-only. > [CmdletBinding()] param([Parameter(Mandatory)][string]$ResourceGroupName, [Parameter(Mandatory)][string]$HostPoolName, [string]$ExpectedImageVersion = '1.0.0', [string]$ExpectedProfileUnc = '', [string[]]$SessionHostComputerNames = @())